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Article in *Journal of Composites for Construction* · April 2013

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Laboratory Characterization and Evaluation of Durability Performance of New Polyester and Vinylester E-glass GFRP Dowels for Jointed Concrete Pavement

Mathieu Montaignu¹; Mathieu Robert²; Ehab A. Ahmed, M.ASCE³; and Brahim Benmokrane⁴

Abstract: This paper presents an experimental study that investigated the physical, mechanical, and durability characteristics of newly developed vinylester- and polyester-based glass-fiber-reinforced polymer (GFRP) dowels through a collaboration research project with the Ministry of Transportation of Québec (MTQ). Durability performance was first evaluated with a preliminary set of conditionings in different solutions. The long-term performance was then assessed under harsh alkaline exposure simulating the concrete environment. The alkaline exposure was achieved by immersing the dowels in an alkaline solution at elevated temperatures to accelerate the effects. Thereafter, the properties were assessed and compared with the unconditioned reference values. The test parameters were (1) type of resin (vinylester and polyester), (2) diameter of GFRP dowels (25.4–44.8 mm), (3) temperature (23, 50, and 60°C), and (4) conditioning time (30, 60, and 180 days). The test results revealed that the vinylester-based GFRP dowels were stable and did not show significant changes in their properties after being subjected to an alkaline solution for 180 days at 60°C. The long-term predictions revealed that the transverse shear-strength retention of vinylester- and polyester-based GFRP dowels would decrease by 10 and 26%, respectively, after 100 years at 10°C. Based on the findings of this research, a new material specification for GFRP dowels was formulated by the MTQ in order to ensure product performance and longevity. DOI: [10.1061/\(ASCE\)CC.1943-5614.0000317](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000317). © 2013 American Society of Civil Engineers.

CE Database subject headings: Concrete pavements; Fiber reinforced polymer; Mechanical properties; Durability; Alkalies; Aging (material); Microstructures; Predictions; Material properties; Dowels.

Author keywords: Concrete; Pavement; GFRP; Dowel; Physical and mechanical properties; Durability performance; Alkaline; Accelerated aging; Microstructural; Prediction; Material specification.

Introduction

Dowel bars are installed at the transverse joints of concrete slabs to reduce the deflections and stresses at the joints while transferring the traffic load (Westergaard 1928). Plain round steel dowels are the most frequently used load-transfer devices in jointed pavement between two adjacent slabs at the joint location. Corrosion of steel dowels in jointed pavement is characterized as the primary source of premature concrete failure by the American Concrete Pavement Association (ACPA). The penetration of chloride-containing deicing salts through the joint initiates the corrosion.

Unlike the United States, which has the Federal Highway Administration, Canada has no single agency responsible for funding pavement construction and rehabilitation or for setting pavement design standards. Pavement design for Canada's primary highway network comes under provincial jurisdiction, while the federal government retains responsibility for national-park roadways. Each agency is free to use whatever design procedure it chooses for pavement design and rehabilitation (Canadian Strategic Highway Research Program 2002). The Ministry of Transportation of Québec (MTQ) has been trying to overcome corrosion-associated problems when steel dowels are used in jointed pavement. More than 350 km of jointed pavement has been built in Québec since 1994. The method commonly employed for more durable pavement has been using epoxy-coated steel dowels. These dowels, however, have evidenced the same performance problems normally associated with surface coating, including voids and damage during transportation and handling. Thus, replacing the steel dowels with noncorroding glass-fiber-reinforced polymer (GFRP) dowels would eliminate the potential for corrosion and related deterioration. On the other hand, few studies have been conducted on the GFRP to investigate their durability performance under real and simulated harsh environmental conditions.

With a main objective of eliminating the potential for corrosion of steel dowels in jointed pavement, MTQ characterized GFRP dowels as alternatives for jointed pavement as the result of a collaboration project with the University of Sherbrooke. This research project aimed at comparing the performance of GFRP dowels in terms of physical, mechanical, and durability properties. The project involved different dowel sizes and composition (resin type) because the polyester-based [unsaturated polyester (UP)] dowels

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Note. This manuscript was submitted on February 25, 2012; approved on June 5, 2012; published online on August 7, 2012. Discussion period open until September 1, 2013; separate discussions must be submitted for individual papers. This paper is part of the *Journal of Composites for Construction*, Vol. 17, No. 2, April 1, 2013. © ASCE, ISSN 1090-0268/2013/2-176-187/\$25.00.

are less expensive than vinylester-based (VE) dowels (about 30% for equivalent diameters). The long-term performance of UP GFRP dowels in alkaline environments (such as moist concrete pavement), however, was a potential issue. Thus, this investigation included both UP- and VE-based dowels. The research project was divided into two main stages. The first stage dealt with the complete characterization and durability evaluation of GFRP dowels manufactured with UP or VE resin. The second stage involved the structural performance of GFRP dowels in jointed pavement under static loading and cyclic loading, and field implementation (Benmokrane et al. 2012). This paper focuses on the first stage of this extended research project, namely, the complete characterization and durability evaluation of UP- and VE-based GFRP dowels under harsh exposure to an alkaline solution simulating a concrete environment. The paper also includes a long-term prediction based on the Arrhenius models to introduce a relative time-shift factor (TSF) for aging VE- and UP-based GFRP dowels.

Research Significance

This paper introduces experimental results on the physical and mechanical characteristics of newly developed VE- and UP-based GFRP dowels. It also assesses and models the long-term mechanical performance of both types in a harsh alkaline environment at high temperatures (accelerated aging). The results contribute to gaining approval for such alternative GFRP dowels in jointed pavements for durable and cost-effective pavement designs. Based on this research, the MTQ issued new material specification for the use of GFRP dowels in jointed concrete pavement in the province of Québec.

Literature Review

FRP composites, mainly based on thermoset polymers and glass or carbon fibers, are being used in concrete infrastructure exposed to harsh conditions. A typical steel dowel bar is more vulnerable to corrosion when used to transfer loads in jointed pavement due to the penetration of water and salts through the joint. When dowel corrosion occurs, freezing could cause significant expansion, followed by fracturing due to curling of the concrete pavement slab. GFRP dowel bars can now be considered as a feasible solution to this problem (Eddie et al. 2001; Porter and Guinn 2002). Engineers and owners, however, have raised concern about the long-term performance of GFRP dowels in highly alkaline environments.

Won et al. (2006) estimated the long-term durability of 25- and 38-mm GFRP dowels manufactured using a VE resin. The GFRP dowels were conditioned at 70°C for 60 days in an alkaline environment consisting of 0.16% $\text{Ca}(\text{OH})_2$, 1% NaOH, and 1.4% KOH with a pH around 12.6, which is close to that of concrete and simulates pore water. Won et al. (2006) used Litherland et al.'s method (1981) to establish a TSF and to assess the durability of GFRP dowels in the Korean environment. The short-beam shear test was used to characterize the degradation of the matrix–fiber interface. No significant degradation of mechanical properties was observed (strength retention over 95%). The short-term accelerated tests indicated that integrity would last over 60 years of service life under Korean road conditions. Thus, longer aging tests and more representative conditioning are needed.

GFRP dowel bars deteriorate when free hydroxyl ions and water molecules diffuse throughout the matrix (Robert et al. 2009). The relative weak adhesion of UP and VE resins could result in serious deterioration when ion-laden water penetrates into the structure. The matrix fails primarily through expansion and

plasticization mechanisms, due to the diffusion of the surrounding solution. In water and alkaline solutions, the deterioration mechanisms of the glass fibers essentially depend on leaching and etching. External alkaline ions leaching into the glass fibers lead to the most important chemical reaction, which dissolves the glass fibers in water. Etching results from hydroxyl ions breaking the Si-O-Si bonding. The concentration and growth of hydration products between individual filaments can also damage the fibers (Murphy et al. 1999). The deterioration at the fiber–matrix interface involves a much more complex mechanism (Chen et al. 2007). The interface is a nonhomogeneous region that is about 1 μm thick, weakly bonded, and most vulnerable to deterioration. Therefore, GFRP degradation is not only due to high pH, but also due to the combination of alkalis, high pH, and moisture. The matrix formed by VE, which contains fewer ester units than UP, is less vulnerable to deterioration by hydroxyl ions. Using cheaper resins, such as UP, could, however, lead to cost-effective structures.

The long-term durability performance of FRP in alkaline environments has been investigated with accelerated aging simulation and predictive models, especially in the case of GFRP bars (Porter et al. 1997; Dejke 2001; Bank et al. 2003; Chen et al. 2006). The predictive models were based on an Arrhenius-type model developed by Litherland et al. (1981). The investigations concluded that the effect of temperature on GFRP-bar durability in alkaline environments could be defined with Arrhenius laws for each temperature difference. These governing factors differ for each product, fiber type, resin, and bar size. In addition, the factors are affected by environmental conditions, such as surrounding solution media, temperature, pH, moisture, and freeze–thaw conditions. Arrhenius-based models implicitly assume that the elevated temperature will only accelerate degradation without affecting its mechanism or introducing other degradation mechanisms. Gerritse (1998) indicated that at least three elevated temperatures were necessary to perform an accurate predication based on an Arrhenius law. Moreover, the measured data should be in continuous time intervals (Robert et al. 2009).

Experimental Program Outline

This experimental investigation was conducted in three phases. *Phase I* included the physical characterization of different diameters of VE- and UP-based GFRP dowels. The physical properties of this phase served as references for the physical properties after conditioning. *Phase II* included the mechanical characterization of different diameters of VE- and UP-based GFRP dowels. The transverse shear strength, flexural strength, flexural modulus of elasticity (stiffness), and the interlaminar shear were determined according to the appropriate test methods. The test results also served as references for residual strength after conditioning. *Phase III* included the preliminary durability evaluation of both the VE- and UP-based GFRP dowels in different chemical solutions and the long-term performance evaluation in alkaline solutions simulating concrete pore solution. Only two of the most common diameters of GFRP dowels were used in this phase (34.9 and 38.1 mm). The GFRP dowels were conditioned by immersion in an alkaline solution simulating the concrete environment for 180 days at different temperatures: 23, 50, and 60°C. The changes in the physical and mechanical characteristics were evaluated by comparing the characteristics of the conditioned GFRP dowels with the reference ones from Phases I and II. The effects of conditioning on the glass-transition temperatures (T_g) and on dowel chemical composition were also evaluated using differential scanning

calorimetry (DSC) and Fourier transform infrared spectroscopy (FTIR), respectively. In addition, the microstructure of both types of GFRP dowels was investigated using scanning electron microscopy (SEM) for both conditioned and unconditioned cases to evaluate changes and/or degradation. *Phase III* included a long-term prediction based on the Arrhenius relation to introduce a relative TSF for aging the VE- and UP-based GFRP dowels.

Materials and Test Specimens

Two different types of GFRP dowels were used in this study (VE and UP based). The GFRP dowels were made of continuous E-glass fibers impregnated in ISO UP or VE resin using the pultrusion process. The physical and mechanical characterizations (Phases I and II) were conducted on VE-based GFRP dowels measuring 25.4, 28.6, 31.8, 34.9, 38.1, 41.3, and 44.5 mm in diameter and UP-based GFRP dowels measuring 25.4, 28.6, 31.8, and 34.9 mm in diameter. The durability performance and the effect of severe conditioning on the characteristics of the GFRP dowels were conducted on two commonly used diameters, namely, 34.9 and 38.1 mm for the VE-based GFRP dowels and 34.9 mm for the UP-based GFRP dowels.

The GFRP dowels were provided in 300- and 1,067-mm lengths so that the transverse shear-strength test and flexural test could be performed according to ACI 440.3R-04 [American Concrete Institute (ACI) 2004] and ASTM D4476 (ASTM 2009), respectively. In addition, some specimens were cut into 150- and 200-mm lengths so that the short-beam shear test could be performed according to ASTM D4475 (ASTM 2008b) on dowels with diameters under or over 31.8 mm, respectively. The characterization test results from Phases I and II served as references for the durability evaluation tests in Phase III.

Testing, Results, and Discussion

Phase I: Physical Characterization

The reference physical properties for the unconditioned UP- and VE-based GFRP dowels were determined according to the test methods of ACI 440.6M-08 (ACI 2008) and CAN/CSA-S807-10 (Canadian Standards Association 2010). Table 1 lists the reference physical properties of the unconditioned UP- and VE-based GFRP dowels.

Table 1 shows that the fiber content of the GFRP dowels was around 80%, while the cure ratio was around 100%, without significant differences between the VE- and UP-based GFRP dowels. The coefficients of thermal expansion between the two dowel types did not differ significantly. A significant difference was observed in the glass-transition temperature (T_g) and moisture uptake. The VE-based GFRP dowels showed a T_g ranging from 118 to 125°C, with a moisture uptake of about 0.07%. On the other hand, the UP-based GFRP dowels evidenced a T_g ranging from 87 to 90°C, with a moisture uptake of about 1.5%, which is 23 times that of the VE-based GFRP dowels. This significant difference may have been related to the delamination and debonding of the fiber-resin interface.

Phase II: Mechanical Characterization

The mechanical characterization included testing of representative GFRP dowels to determine their transverse shear strength [ACI 440.3R-04 (ACI 2004)], interlaminar shear (short-beam test) [ASTM D4475 (ASTM 2008b)], flexural strength, and flexural modulus of elasticity [ASTM D4476 (ASTM 2009)]. Fig. 1 shows the set up for mechanical characterization of the GFRP dowels. Table 2 lists the reference mechanical properties of unconditioned

Table 1. Physical Properties of the Reference GFRP Dowels

Property	VE-based GFRP dowels							UP-based GFRP dowels			
	VE25	VE29	VE32	VE35	VE38	VE41	VE45	UP25	UP29	UP32	UP35
Specimen											
Diameter (mm)	25.4	28.6	31.8	34.9	38.1	41.3	44.5	25.4	28.6	31.8	34.9
Fiber weight content (%)	81.5	81.3	80.5	80.7	80.6	80.9	80.3	80.6	80.6	79.6	79.8
Cure ratio (%)	100	100	100	100	100	100	100	99.8	99.9	99.9	99.9
T_g (°C)	118	119	124	124	123	121	125	87	83	84	90
Moisture uptake (%)	0.07	0.08	0.05	0.06	0.07	0.07	0.07	1.46	1.59	1.50	1.53
Relative density	2.28	2.22	2.14	2.12	2.14	2.12	2.08	2.18	2.20	2.13	2.11
CTE long. ($\times 10^{-6} \text{°C}^{-1}$)	7.5	7.6	7.7	7	7.6	7.6	7.8	7.8	6.9	7.6	6.8
CTE transverse ($\times 10^{-6} \text{°C}^{-1}$)	25.4	25.9	27.1	23.5	24	27.6	26.6	23.2	23.9	26.5	21

Note: CTE = coefficient of thermal expansion.

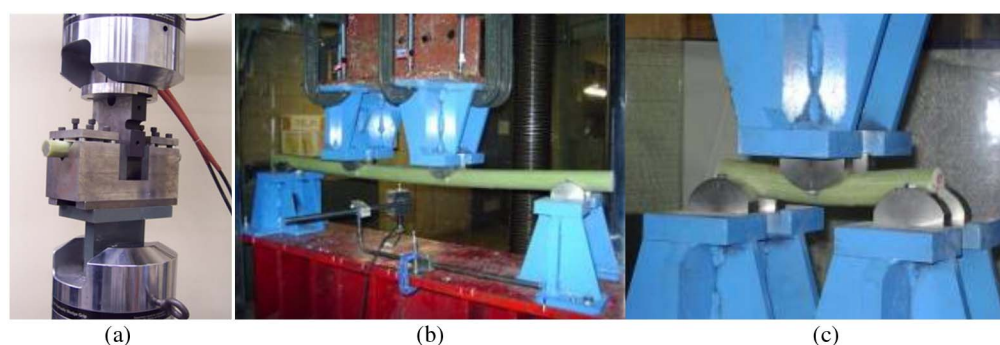


Fig. 1. Experimental setup for mechanical characterization: (a) transverse shear test setup; (b) four-point flexural test; and (c) interlaminar shear test

Table 2. Mechanical Properties of the Reference GFRP Dowels

Property	VE-based GFRP dowels						UP-based GFRP dowels				
	VE25	VE29	VE32	VE35	VE38	VE41	VE45	UP25	UP29	UP32	UP35
Specimen											
Diameter (mm)	25.4	28.6	31.8	34.9	38.1	41.3	44.5	25.4	28.6	31.8	34.9
Transverse shear strength (MPa)	168 ± 2.4	194 ± 2.3	160 ± 3.3	184 ± 1.8	173 ± 3.2	197 ± 3.7	182 ± 2.9	154 ± 1.5	166 ± 3.7	151 ± 1.3	164 ± 4.4
Short-beam shear strength (MPa)	53.2 ± 2.3	59.9 ± 2.0	60.1 ± 2.4	61.1 ± 0.2	53.9 ± 1.5	58.3 ± 1.5	60.2 ± 1.3	29.1 ± 0.7	34.8 ± 1.3	40.1 ± 1.7	36.7 ± 2.2
Four-point flexural strength (MPa)	1,324 ± 63	1,213 ± 62	1,165 ± 37	1,210 ± 50	1,077 ± 61	1,015 ± 22	1,040 ± 32	1,081 ± 72	788 ± 49	839 ± 33	759 ± 36
Flexural modulus (stiffness) (GPa)	54.1 ± 0.7	53.4 ± 1.5	52.6 ± 0.7	50.3 ± 0.5	51.6 ± 0.8	50.2 ± 1.0	49.3 ± 0.7	51.1 ± 1.5	50.2 ± 1.1	51.6 ± 1.0	49.5 ± 0.8

Note: VE25 = VE-based GFRP dowels measuring 25.4 mm in diameter.

UP- and VE-based GFRP dowels. The following sections provide brief descriptions and interpretation of the results for each test.

Transverse shear-strength test. Transverse shear is the major structural solicitation on dowels in jointed pavement. Transverse shear tests were conducted to characterize the GFRP dowels according to ACI 440.3R-04 (ACI 2004) test method B.4. For comparison, 28.6- and 38.1-mm epoxy-coated steel dowels were also tested. The set up consisted of a 230 × 100 × 110-mm steel base equipped with lower blades spaced at 50 mm face to face, allowing for the double transverse shear failure of the specimen caused by an upper blade, as shown in Fig. 1(a). It is worth mentioning that the current standard for transverse shear testing is ASTM D7617 (ASTM 2011b).

Six unconditioned specimens 300 mm in length were tested for each type and diameter of GFRP dowel under laboratory conditions using a MTS 810 testing machine equipped with a 500-kN load cell. A displacement-controlled rate of 1.5 mm/min was used during the test, which yielded between 30 and 60 MPa/min until specimen failure. The transverse shear strength was calculated from Eq. (1) as follows:

$$\tau_u = \frac{P_s}{2A} \quad (1)$$

where τ_u = transverse shear strength (MPa); P_s = failure load (N); and A = cross-sectional area of the dowel (mm²).

Table 2 shows that the transverse shear strength of the VE-based GFRP dowels ranged from 160 to 184 MPa, with an average of 180 MPa. The UP-based GFRP dowels had transverse shear strength ranging from 151 to 164 MPa, with an average of 159 MPa, which is 0.88 that of the VE-based GFRP dowels. Transverse shear strength results mainly from the resin, although the fiber and the fiber–resin interface play a role. The results indicate that the UP resin led to lower transverse shear strength than the VE resin. The ratios between the average shear strengths of the VE- and UP-based GFRP dowels and that of epoxy-coated steel dowels (240 MPa) were 0.75 and 0.66, respectively.

Four-point flexural test. The deformation of the dowel bar crossing the crack has both a shearing and a flexural component. Therefore, flexural testing of GFRP dowels was conducted according to the four-point flexural test method developed by Zhang et al. (2006) and was consistent with ASTM D4476 (ASTM 2009). This method is more appropriate for pultruded bars, allowing the failure to occur in the weakest zone of the material, in contrast to the three-point test, which leads to failure under the loading nose. The tests were conducted on 1,067-mm-long specimens over a simply supported span equal to 16 times the dowel diameter, as shown in Fig. 1(b). Six unconditioned specimens were tested as references for each type and diameter under laboratory conditions on a 1,800-kN Baldwin testing machine. The specimens were tested under a displacement-controlled rate of 60 mm/min, which gave between 250 and 350 MPa/min until specimen failure due to one of the following: (1) tension, (2) compression, or (3) maximum strain of 5%. The flexural strength was calculated from Eq. (2).

$$\sigma_u = \frac{2PL}{3\pi R^3} \quad (2)$$

where σ_u = flexural stress in the outer fibers at midspan (MPa); P = failure load (N); L = support span (mm); and R = dowel bar radius (mm).

The flexural modulus of elasticity (stiffness) was calculated using Eq. (3) for a chosen P and a corresponding deflection Y measured with a linear variable-displacement transducer.

$$E = \frac{23L^3}{1,296(\pi R^4/4)} \frac{P}{Y} \quad (3)$$

The maximum outer fiber strain can be calculated from Eq. (4).

$$\varepsilon_{\max} = \frac{\sigma_u}{E} \quad (4)$$

Table 2 provides the four-point flexural strength of the tested GFRP dowels and their flexural moduli of elasticity. The VE-based GFRP dowels showed a flexural strength ranging from 1,015 to 1,324 MPa, with an average of 1,149 MPa, while the UP-based GFRP dowels recorded flexural strength ranging from 759 to 1,081 MPa, with an average of 867 MPa. Because the relationship is evident for both types of GFRP dowels tested, the larger the diameter, the lower the flexural strength. On the other hand, there were no significant differences between the flexural modulus of elasticity of both GFRP dowel types, which was close to 50 GPa.

Short-beam shear test. GFRP dowel bars are manufactured by a pultrusion process in which the fibers are arranged unidirectionally and bonded with the polymer matrix. Therefore, horizontal stresses would be more conducive to inducing interface degradation than transverse shear stresses (Park et al. 2008). The short-beam shear test was conducted on six specimens of each GFRP dowel type and diameter according to ASTM D4475 (ASTM 2008b). This test was conducted to get the interlaminar shear strength, which is governed by the fiber–matrix interface. The tests were carried out using a 500-kN MTS 810 testing machine. The distance between the shear planes was set to three times the nominal diameter of the GFRP dowels. Fig. 1(c) is a picture of the test setup. A displacement-controlled rate of 1.3 mm/min was employed during the test. The applied load was recorded with a computer-monitored data-acquisition system.

The interlaminar shear strength, S_u , of GFRP bars was calculated from Eq. (5) as follows:

$$S_u = \frac{0.849P}{d^2} \quad (5)$$

where S_u = interlaminar shear strength (MPa); P = shear failure load (N); and d = bar diameter (mm).

The short-beam shear tests revealed that the interlaminar shear strength of the UP-based GFRP dowels was very low compared with that of the VE-based ones. The VE-based GFRP dowels provided an average short-beam shear strength of 58.1 MPa, compared with 35.2 MPa for the UP-based GFRP dowels. The average short-beam strength (which is an interface-governed property) of the UP-based GFRP dowels was equal to 0.61 of that of the VE-based GFRP dowels. The interface between the fibers and the UP resin was not as strong as the interface between the fibers and the VE resin. The presence of the secondary hydroxyl groups in the VE molecule, which can interact with the hydroxyl groups present on the surface of glass fibers, and improve wetting and bonding of the resin to the reinforcement, accounts for these results (Boinard et al. 2000). Besides, there was no clear effect for the diameter on the interlaminar shear of the GFRP dowels.

Phase III: Durability Study and Long-Term Performance Evaluation

Preliminary durability study. A preliminary durability study was performed on 35-mm VE- and UP-based GFRP dowels before evaluating the long-term performance of GFRP dowels subjected to an alkaline solution. This preliminary study was performed to evaluate the durability of GFRP dowel bars in different chemical solutions. Three different conditions were prepared to match those

in actual concrete pavement structures. The environments for the tests were (1) tap water with a pH of 7.4 used as a reference solution, according to ASTM D2247 (ASTM 2011a); (2) a solution of CaCl_2 by weight with a pH of 7.65 used to simulate aggressive deicing salts harsher than NaCl, according to the ASTM D1141 (ASTM 2008a); and (3) a $\text{Ca}(\text{OH})_2$ saturated solution with a pH of 12.65 used to simulate a concrete pore-water solution saturated with calcium ions. Six 241-mm-long specimens of each type of resin and diameter were immersed in a PVC tray. The plastic containers were tightly closed to avoid excessive evaporation of water during the conditioning. The water level and pH were kept constant throughout the study. The specimens were completely immersed at 23°C for 90 days. The six GFRP dowels were then subjected to short-beam shear testing to compare their residual interlaminar shear strength with that of the reference specimens.

The retention of interlaminar shear strength after immersion of the UP- and VE-based specimens in tap water, CaCl_2 solution, and $\text{Ca}(\text{OH})_2$ solution is provided in Fig. 2. Fig. 2 shows that the VE-based GFRP dowels presented very high stability and durability in the different chemical solutions, as evidenced by no significant loss of properties after 90 days of conditioning at 23°C. On the other hand, the tap water, $\text{Ca}(\text{OH})_2$ solution, and CaCl_2 solution led to a loss of around 12% of the UP-based dowel's interlaminar shear strength, regardless of aging solution pH.

Conditioning of the GFRP dowels for long-term performance evaluation. Further to the preliminary durability study, accelerated aging tests were conducted according to the Dowel Bar Team 2-recommended FRP Dowel Bar Test Protocol proposed by the Market Development Alliance of the Composite Institute (Market Development Alliance 1998). This empirical method employs accelerated conditioning techniques for 180 days to validate the satisfactory performance of GFRP dowels in jointed pavement in real service with minimal maintenance for 30–40 years. Conditioning of the GFRP dowels included the combination of exposure to a harsh alkaline environment and elevated temperature. High temperature during immersion in an aqueous media (alkaline solution) will accelerate effects. A cement-extract solution—simulating the alkaline environment of concrete—was prepared by mixing commercial type 30 portland cement (highest rate of CaO) with tap water to get a pH of 12.65, measured at the beginning of the test and maintained constant throughout the conditioning period with $\text{Ca}(\text{OH})_2$ and cement. The GFRP dowels were immersed in the alkaline solution at three different temperatures: 23, 50, and 60°C. Once the solution reached the prescribed temperature, the conditioning time was started. Robert et al. (2009) reported that the increase

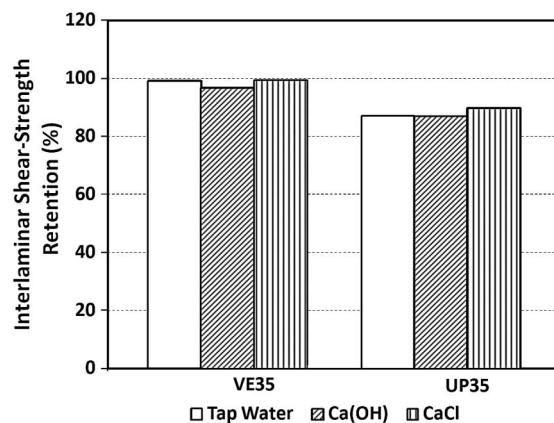


Fig. 2. Interlaminar shear-strength retention after aging in different chemical solutions

Table 3. Transverse Shear and Flexural Strengths of the Conditioned GFRP Dowels

Period of immersion (days)	Temperature (°C)	Average transverse shear strength (MPa) (retention)			Average flexural strength (MPa) (retention)		
		VE35 dowel	VE38 dowel	UP35 dowel	VE35 dowel	VE38 dowel	UP35 dowel
0	23	184 ± 1.8	173 ± 3.2	164 ± 4.4	1,210 ± 50	1,077 ± 61	759 ± 36
30	23	182 ± 2.7 (0.99)	178 ± 1.2 (1.03)	153 ± 1.9 (0.93)	1,202 ± 31 (0.99)	1,080 ± 26 (1.00)	714 ± 26 (0.94)
	50	180 ± 2.4 (0.98)	176 ± 2.5 (1.02)	150 ± 2.7 (0.91)	1,200 ± 38 (0.99)	1,075 ± 36 (1.00)	690 ± 16 (0.91)
	60	178 ± 3.7 (0.97)	171 ± 1.9 (0.99)	148 ± 3.7 (0.90)	1,191 ± 76 (0.98)	1,065 ± 56 (0.99)	648 ± 13 (0.85)
60	23	180 ± 2.0 (0.98)	182 ± 1.9 (1.05)	143 ± 3.6 (0.87)	1,195 ± 27 (0.99)	1,085 ± 20 (1.00)	669 ± 21 (0.88)
	50	176 ± 4.2 (0.96)	178 ± 1.8 (1.03)	136 ± 4.0 (0.83)	1,190 ± 37 (0.98)	1,077 ± 36 (1.00)	622 ± 18 (0.82)
	60	172 ± 1.9 (0.93)	169 ± 0.9 (0.98)	133 ± 4.2 (0.81)	1,183 ± 74 (0.98)	1,058 ± 54 (0.98)	606 ± 7 (0.80)
180	23	179 ± 1.9 (0.97)	177 ± 2.1 (1.02)	138 ± 2.8 (0.84)	1,190 ± 18 (0.98)	1,087 ± 70 (1.01)	612 ± 23 (0.81)
	50	173 ± 3.1 (0.94)	170 ± 0.7 (0.98)	131 ± 1.5 (0.80)	1,179 ± 19 (0.97)	1,058 ± 37 (0.98)	555 ± 25 (0.73)
	60	168 ± 3.7 (0.91)	158 ± 3.3 (0.91)	125 ± 1.6 (0.76)	1,154 ± 53 (0.95)	1,049 ± 12 (0.97)	498 ± 13 (0.66)

in reaction rate of degradation was almost linear between room temperature and 50°C, whereas, at higher temperatures (over 60°C), the increase in reaction/degradation rate was exponential. Therefore, to avoid any thermal degradation, the maximum conditioning temperature used in this study was 60°C. The GFRP dowels were placed in wooden containers made watertight with high-density polyethylene film and hermetically sealed to prevent excessive evaporation and changes in solution pH. PVC sticks were used to maintain specimen separation so that the solution could cover the entire GFRP dowel surface [ASTM C581 (ASTM 2003)]. The containers were placed in three environmental chambers adjusted to the prescribed temperatures (23, 50, and 60°C) under isothermal conditions. The GFRP dowels were weighed and their diameters were measured throughout the conditioning period to monitor water absorption and eventually characterize the mass and diameter changes. The observation did not show any change in diameter during the conditioning periods. Six specimens of each diameter and resin type were removed from the solution and tested to determine their transverse shear strength and flexural strength after 30, 60, and 180 days at 23, 50, and 60°C of conditioning to measure the mechanical properties of the aged GFRP dowels. The experimental data were used to estimate the simulated long-term exposure to alkaline solution on the properties and performance of GFRP dowels and to characterize any degradation mechanisms in the GFRP dowels.

Transverse shear strength of conditioned GFRP dowels. Table 3 shows the transverse shear strength of both the VE- and UP-based dowels after conditioning in an alkaline solution. Table 3 shows the retention of the transverse shear strength of the GFRP dowels after 30, 60, and 180 days of immersion in an alkaline solution at 60°C. Table 3 indicates that the VE-based dowels were slightly affected by accelerated aging with transverse shear-strength retention of more than 90% after 180 days at 60°C in an alkaline solution. The UP-based dowels showed transverse shear-strength retention above 75%. The UP resin was affected by conditioning in an alkaline solution more than the VE one.

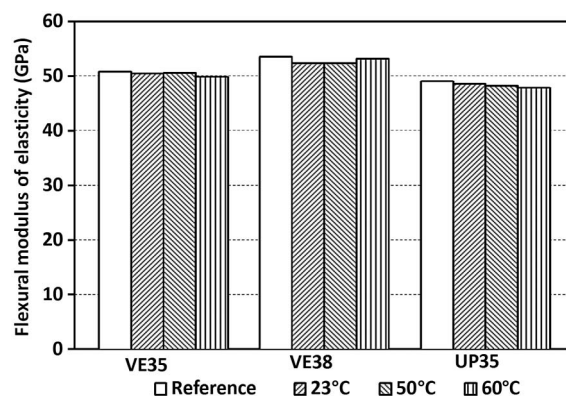
Flexural strength of conditioned GFRP dowels. Table 3 shows the flexural properties of both the VE- and UP-based GFRP dowels after conditioning as well as the flexural strength retention. The VE-based dowels showed a high level of flexural-strength retention. Even after 180 days of conditioning in an alkaline solution at 60°C, the retained flexural strength was equal to 95% of the initial flexural strength. The UP-based dowels showed flexural-strength retention above 65%. The elastic behavior of all specimens was retained until flexural failure. The VE-based dowels failed under different modes (compression, balanced, or tension) and the UP-based dowels evidenced tensile rupture. This observation confirmed that the bond between the glass fiber and polymer

was better in the VE resin than between the glass fibers and UP resin, which led to failure at the fiber–matrix interface.

Flexural modulus of elasticity (stiffness) of the conditioned GFRP dowels. Fig. 3 shows the change in the flexural modulus of elasticity (stiffness) of the aged VE- and UP-based GFRP dowels after immersion in an alkaline solution for 180 days at various temperatures. After 180 days of immersion, the loss of flexural modulus of elasticity was negligible for both the VE- (2%) and UP-based (5%) dowels. The results also confirmed that the aged VE-based dowels were not significantly affected by the higher temperature or exposure to an alkaline solution. These results also showed that the flexural elastic modulus of elasticity was not significantly affected by aging in an alkaline solution for 180 days. This result can be explained by the fact that modulus of elasticity is primarily a fiber-governed property and that the glass fibers were not degraded by the alkaline solution used.

Microstructural changes in the conditioned GFRP dowels. SEM observations and image analysis were performed to observe the microstructure of both the VE- and UP-based dowels specimens before and after aging in an alkaline solution for 180 days at 60°C. All specimens observed under SEM were cut, polished, and coated with a thin layer of gold/palladium in a vapor-deposit process. Thereafter, microstructural observations were performed on a JEOL JSM-840A SEM to investigate the potential degradation of polymer matrix, glass fibers, or interfaces, if any. Microscopic observation of the mode of failure was also performed on specimens after flexural testing to identify any changes in the mode of failure due to resin-type or alkaline conditioning.

SEM observations of the external surface of the GFRP dowels were conducted to investigate the surface deterioration of the polymer matrix after conditioning in an alkaline solution. Fig. 4

**Fig. 3.** Elastic modulus of the reference and conditioned GFRP dowels

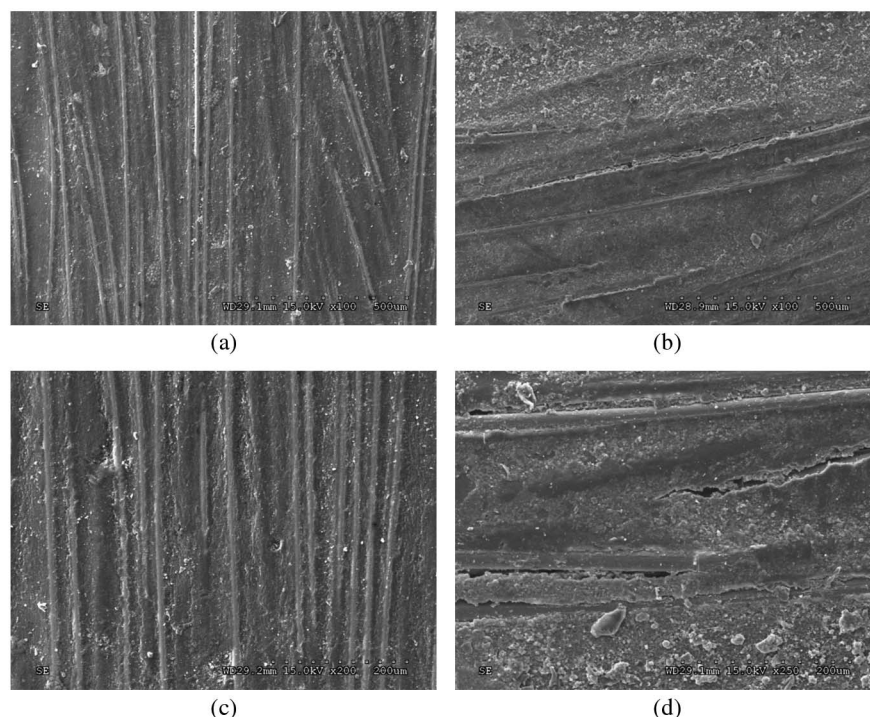


Fig. 4. Micrographs of the dowel surface: (a) 34.9-mm VE reference dowel; (b) 34.9-mm UP reference dowel; (c) 34.9-mm VE dowel aged in an alkaline solution for 180 days at 60°C; (d) 34.9-mm UP dowel aged in an alkaline solution for 180 days at 60°C

presents micrographs of the surface of the reference dowels as well as the aged VE- and UP-based dowels. It shows no degradation of the VE matrix after 180 days at 60°C aging in an alkaline solution. The pore number and size remained unchanged; the matrix was intact with no cracking or microcracking. On the other hand, the conditioning led to local surface degradation in the UP-based dowels. The matrix was more porous and some cracks appeared, which could lead to increased penetration of the surrounding solution and a higher diffusion of hydroxyl ions in the polymer matrix. No dowel size or weight changes were noticed after accelerated aging in an alkaline solution.

SEM observations were also performed on reference and aged dowels after flexural testing to investigate the mechanisms of failure at the fiber–matrix interface. Fig. 5 presents micrographs of the fiber–matrix interface after failure of the reference and conditioned GFRP dowels in an alkaline solution for 180 days at 60°C. The VE-based dowels presented more residual matrix on the fibers after failure compared with the UP-based dowels. This observation corroborates the different modes of failure observed in the VE-based dowels (compression, balanced, or tension) and characterizes the higher bond of the VE resin compared with that of the UP resin, which led to failure at the fiber–matrix interface. The micrograph of the UP-based dowels showed smooth fibers after debonding and failure, which indicates weaker adhesion of the UP resin, due to the presence of weak ester groups, and the easier delamination of fibers at the interface. This observation is in accordance with the tension mode of failure observed for the UP-based dowels.

Fig. 6 presents micrographs of the internal microstructure of both the reference and GFRP dowels conditioned in an alkaline solution for 180 days at 60°C. The visual and microstructural observations showed no significant damage to the surface of the VE-based dowel after 180 days of immersion in an alkaline solution at the highest temperature (60°C). The fiber–matrix interface and the microstructure demonstrated, in general, that the conditioning of the VE-based dowel in an alkaline solution did not affect dowel

microstructural properties [Fig. 6(c)]. As for the UP-based dowels, Fig. 6(b) shows that some delamination occurred initially, which can explain the higher moisture uptake measured at saturation compared with that of the VE-based dowels. The micrograph in Fig. 6(d), taken near the surface, shows the degradation at the fiber–polymer matrix interface, which accounts for the lower interface-governed mechanical strengths evidenced by the UP-based dowels. These observations should, however, be considered carefully due to the local effect of the degradation near the surface.

Glass-transition temperature of conditioned GFRP dowels. DSC is used to obtain information about the thermal behavior and characteristics of polymeric materials and composites, such as glass-transition temperature (T_g), melting point, curing process, crystallinity, thermal stability, and relaxation. In this study, 12–15-mg specimens from both unconditioned and conditioned specimens were sealed in aluminum pans and analyzed in a DSC Q10 (TA Instruments) equipped with a refrigerated cooling system. Analysis was conducted in modulated DSC mode. Specimens were heated from 25 to 195°C at a rate of 5°C/min. Glass-transition temperature was determined by DSC for each specimen in accordance with ASTM D3418 (ASTM 1999). Two scans were performed for each specimen. The first scan helped to determine the difference in T_g between the reference and conditioned specimens. A decrease in T_g for the conditioned specimens is indicative of a plasticizing effect or chemical degradation. Low T_g of conditioned specimens after the second scan is indicative of a nonreversible chemical degradation, such as hydrolysis.

Table 4 presents the T_g values before and after aging in an alkaline solution for 180 days at 60°C. The VE-based dowels displayed no significant effects. The UP-based dowels were affected by the aging, showing a loss of T_g of 6% after conditioning, with no change in diameter. The T_g drop in the second run can be explained by a nonreversible chemical reaction, such as hydrolysis. This corroborates the observed mechanical losses and the local degradation of the matrix structure revealed by SEM (Figs. 4 and 6).

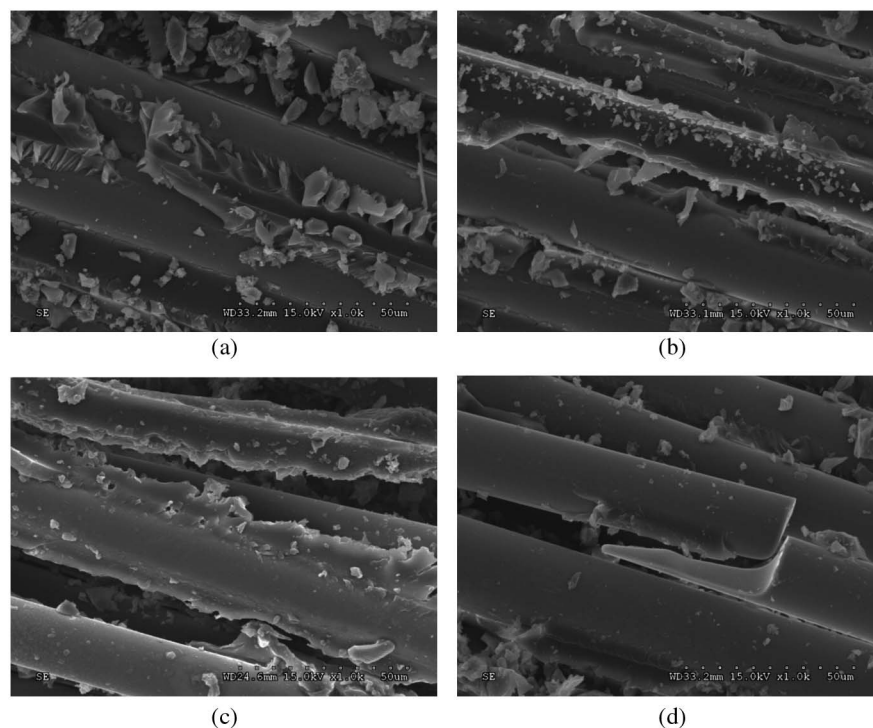


Fig. 5. Micrographs at the fiber/matrix interface after failure: (a) VE reference dowel; (b) UP reference dowel; (c) 34.9-mm VE dowel aged in an alkaline solution for 180 days at 60°C; (d) 34.9-mm UP dowel aged in an alkaline solution for 180 days at 60°C

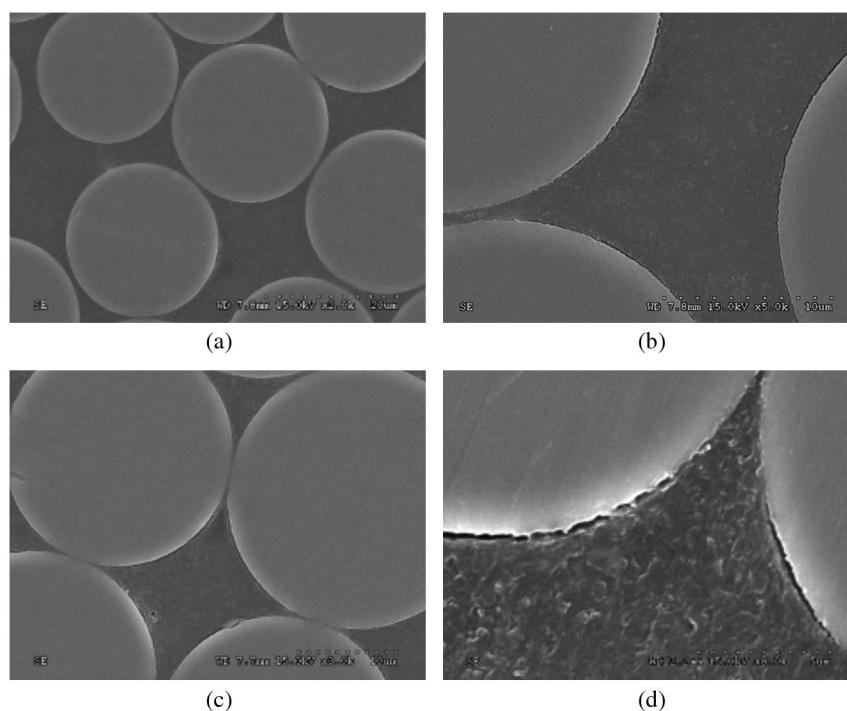


Fig. 6. Micrographs at the fiber-matrix interface before mechanical tests: (a) VE reference dowel; (b) UP reference dowel; (c) 34.9-mm VE dowel aged in an alkaline solution for 180 days at 60°C; (d) 34.9-mm UP dowel aged in an alkaline solution for 180 days at 60°C

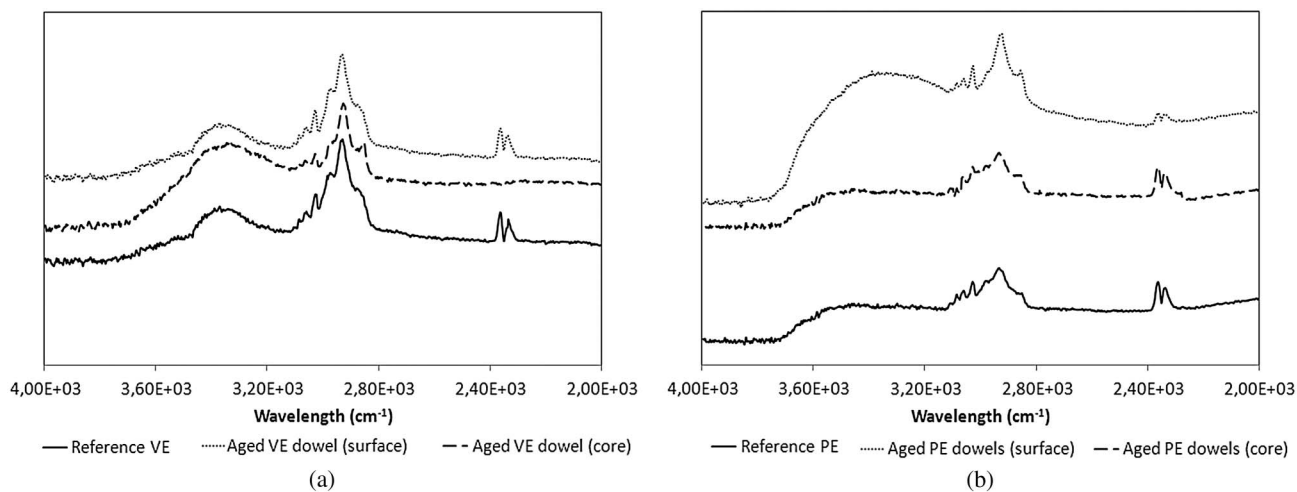
Chemical changes in the conditioned GFRP dowels. FTIR was used to identify any chemical change/degradation of the GFRP dowels after conditioning. FTIR spectra were recorded using a Nicolet Magna 550 spectrometer equipped with an attenuated total reflectance device. Fifty scans were routinely acquired with an

optical retardation of 0.25 cm to yield a resolution of 4 cm^{-1} . The 25.4-mm-diameter dowels were investigated after 180 days of conditioning at 60°C.

Fig. 7 shows the FTIR analysis of the unconditioned dowels and dowels aged in an alkaline solution for 180 days at 60°C.

Table 4. T_g of the Reference and Aged GFRP Dowels

Dowel status	Temperature (°C)	Duration (days)	Glass-transition temperature, T_g					
			Run 1 (°C)			Run 2 (°C)		
			VE35 dowel	VE38 dowel	UP35 dowel	VE35 dowel	VE38 dowel	UP35 dowel
Unconditioned	—	—	123	124	90	124	123	90
Conditioned in alkaline solution	60	180	122	123	84	123	122	85

**Fig. 7.** FTIR spectra for unconditioned and aged samples: (a) vinylester-based dowel and (b) polyester-based dowel

The most interesting region of the FTIR spectra is located between 3,300 and 3,600 cm^{-1} , which corresponds to the stretching mode of the hydroxyl groups in the VE and UP resins. Hydrolysis results in the formation of new hydroxyl groups, with corresponding increases in the infrared band. Changes in the peak intensity are quantified by determining the ratio of the OH^- peak to the resin's carbon-hydrogen stretching peak, which is not affected by conditioning. The experimental ratios of the OH peak to the carbon-hydrogen stretching peak of the core and surface of the VE-based dowels immersed in an alkaline solution for 180 days at 60°C were 0.44 and 0.54, respectively, compared with 0.49 for unconditioned samples. The hydroxyl peak did not show any significant changes. This observation led to the conclusion that the VE resin did not degrade chemically during the immersion of the dowels in an alkaline solution at 60°C for 180 days. On the other hand, the experimental ratios for the core and surface of the UP-based dowels immersed in an alkaline solution for 180 days at 60°C were 0.50 and 0.87, respectively, compared with 0.48 for unconditioned samples. These results led to the conclusion that chemical degradation of the polymer occurred at the dowel surface, which was in direct contact with the solution during immersion. This observation explains the losses in mechanical properties of the UP-based dowels. It can be concluded that the thickness of the degraded polymer matrix is related to moisture absorption and that the degradation is a time-dependent mechanism.

Moisture uptake at saturation of the conditioned GFRP dowels. The moisture uptake at saturation of the reference samples was determined before and after conditioning according to ASTM D570 (ASTM 2010). Three 75-mm-long specimens for each type of GFRP dowel were cut, dried, and weighed. They were then immersed in water at 50°C for 3 weeks. The samples were periodically

removed from the water, surface dried, and weighed. The water content as a percentage of weight was calculated with Eq. (6).

$$W = 100 \times (P_s - P_d) / P_d \quad (6)$$

where P_s and P_d = weights of the bar in the saturated and dried states, respectively. The percentage of moisture uptake was calculated. The gain in mass was corrected to account for possible specimen mass loss due to various dissolution phenomena—such as hydrolysis—during the aging process. This correction was achieved by completely drying the immersed specimens in an oven at 100°C for 24 h and comparing their masses to their initial masses.

The moisture uptake ratios at saturation for the reference specimens were 0.06, 0.07, and 1.35% for VE35, VE38, and UP35, respectively. While these ratios at saturation for the aged specimens were 0.06, 0.07, and 1.63% for VE35, VE38, and UP35, respectively. This indicates no change of moisture uptake in either the VE- or UP-based GFRP dowels. Reference values remained strictly equal to those resulted after aging. In spite of the slight UP-resin degradation characterized by DSC analysis and SEM observations (Fig. 4), aging had no effect on moisture uptake in the UP-based GFRP dowels.

Arrhenius Relation and Long-Term Predictions

Bank et al. (2003) employed the Arrhenius concept for predicting the long-term behavior of GFRP dowels. Eq. (7) expresses the Arrhenius relation, in terms of the degradation rate (Nelson 1990).

$$k = A e^{-\frac{E_a}{RT}} \quad (7)$$

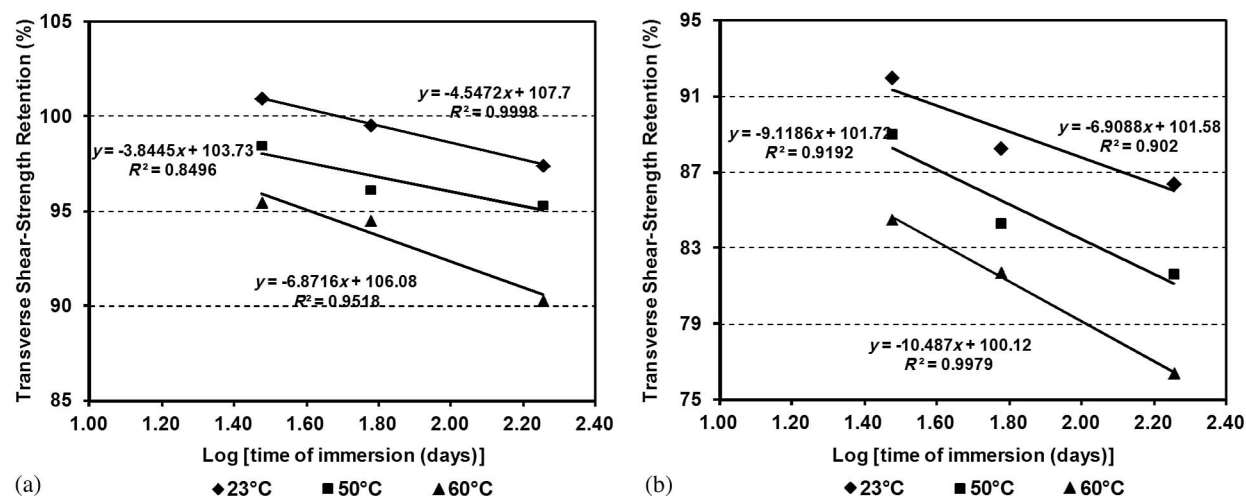


Fig. 8. Arrhenius plots for (a) vinylester-based dowel and (b) polyester-based dowel

where k = degradation rate; A = constant relative to the material and degradation process; E_a = reaction's energy of activation; R = universal gas constant; and T = temperature in °C. The primary assumption with this model is that only one dominant degradation mechanism of the material operates during the reaction and that this mechanism will not change with time and temperature during the exposure (Chen et al. 2007). Only the rate of degradation will be accelerated as the temperature increases. Dowel mechanical properties at different aging temperatures and periods can lead to a long-term extrapolation of material properties in pavement structures.

Dejke (2001) used another approach to generate a relative TSF. He proposed using the TSF to transform the time in the accelerated test to actual service lives for GFRP reinforcement. Because the time for a certain reaction to take place must be proportional to the inverse of the rate of reaction, Dejke (2001) proposed determining the TSF in accordance with Eq. (8).

$$\text{TSF} = \frac{t_1}{t_2} = e^{B(\frac{1}{T_1+273.15} - \frac{1}{T_2+273.15})} \quad (8)$$

where T_1 and T_2 = exposure temperatures (°C); and t_1 and t_2 = times required to obtain a certain level of decrease in mechanical property at temperatures T_1 and T_2 , respectively. The TSF is sensitive to activation energy, and a good estimate of E_a is needed to generate a reasonable TSF.

According to the procedure proposed by Bank et al. (2003), the natural logarithm of time to reach a set of levels of normalized performances versus $1/T$, expressed as the inverse of absolute temperature (1,000/K), was used to predict the service life at 10°C, which represents the mean annual temperature (MAT) in Montréal, Canada (Benmokrane et al. 2008). A coefficient of determination (R^2) value close to 1 was desired. ASTM procedures, however, recommend a minimum value of 0.80 for acceptability; the obtained R^2 values were between 0.96 and 0.99. The Arrhenius plot (Fig. 8) can be used to extrapolate the service lifetime necessary to reach the established transverse shear-strength retention levels (PR) for any temperature. Consequently, predictions were made for transverse shear-strength retention as a function of time for immersion at 10°C. The general relation between the PR and predicted service life is shown in Fig. 9. The VE-based dowels showed very high durability in a concrete environment. In fact, the predicted service life of the VE-based dowels immersed in an alkaline solution at an isothermal temperature of 10°C to reach a PR of less than 85%

can be considered infinite. On the other hand, the prediction curves for the UP-based dowels predicted transverse shear-strength retention of 75% after a service life of 75 years. Table 5 presents the transverse shear-strength retention after 50, 75, and 100 years of service at MATs of 10 and 25°C. Table 5 indicates that, even after a service life of 100 years, which corresponds to the maximum expected service life, the shear-strength retention was still 90 and 74% for the VE- and UP-based dowels, respectively, at 10°C. At 25°C, the shear-strength retention was still 90 and 70% for the VE- and UP-based dowels, respectively.

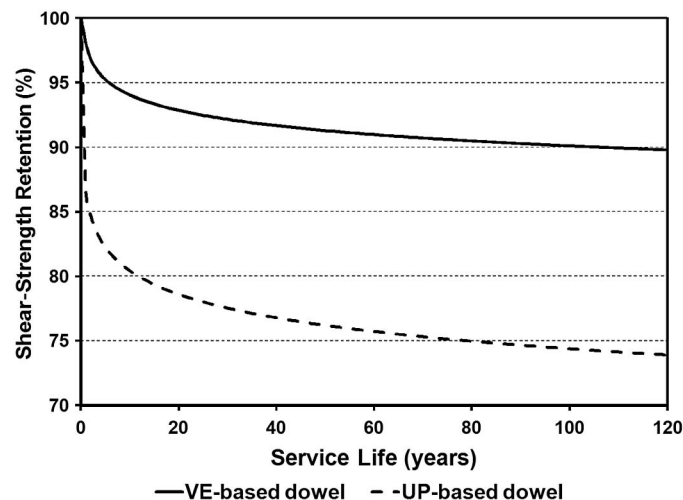


Fig. 9. Long-term predictions for the 34.9-mm dowels

Table 5. Transverse Shear-Strength Retention at Different Service-Life Periods at MATs of 10 and 25°C

Service life (Years)	Transverse shear-strength retention (%)			
	VE-based dowel		UP-based dowel	
	10°C	25°C	10°C	25°C
50	91	87	76	72
75	91	87	75	71
100	90	90	74	70

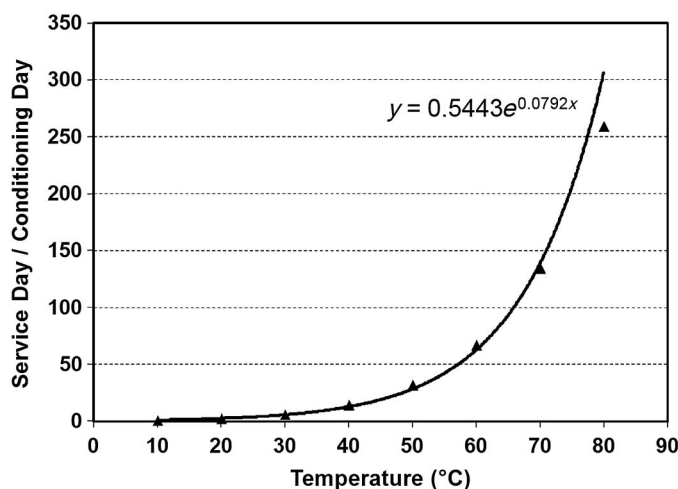


Fig. 10. TSF for shear-strength retention of the UP-based GFRP dowels for equivalent service time at 10°C

In this research the DeJke's approach (DeJke 2001) was used to generate a relative TSF for aging in an alkaline solution. Eq. (8) was used to calculate the TSF for the UP dowels. The parameters t_1 and t_2 were determined for strength retention of 90% at 50 and 60°C, respectively. Fig. 10 provides the TSF curves for the UP-based dowels, where $y = N/C$, where N = equivalent service days at 10°C and C = number of days of conditioning. According to these predictions, 1 day of conditioning at 60°C corresponds to 67 service days under shear load. Therefore, the 180 days of conditioning used in this study leads to a shear-strength retention of 75%, corresponding to a service life of more than 33 years. Because the VE-based dowels showed very slight damage from the accelerated aging tests, it was not possible to generate a reasonable TSF. On the other hand, ACPA estimated that steel dowel bars can fail in 7–15 years, depending on design and location, whereas the concrete highway slab itself can easily perform for 35–40 years (Porter 2002).

Summary and Conclusions

Based on the results of this study, the following conclusions concerning the GFRP dowels tested can be drawn:

1. The major differences in the physical properties between the VE- and UP-based GFRP dowels were glass-transition temperature, T_g , and moisture uptake. The moisture uptake was 23 times that of the VE-based GFRP dowels, which is possibly related to the delamination debonding of the fiber–resin interface.
2. The mechanical properties of the VE-based dowels were higher than those of UP-based dowels, in particular, the interlaminar shear strength of the VE-based dowels, which is governed by the fiber–matrix interface. The average interlaminar shear strength (which is an interface-governed property) of the UP-based GFRP dowels was equal to 0.61 that of the VE-based GFRP dowels.
3. The UP-based dowels were affected by tap water (pH: 7.4), $\text{Ca}(\text{OH})_2$ solution (pH: 12.65), and CaCl_2 solution (pH: 7.65), with a loss of around 12% of the interlaminar shear strength, regardless of the pH of the aging solution.
4. The VE-based dowels showed very high durability after 180 days of accelerated aging in an alkaline solution (pH: 12.65) at 60°C. Strength retentions of 90 and 95% were measured in transverse shear strength and flexural strength,

Table 6. New Material Specification for GFRP Dowels Formulated by the MTQ

Property	Limit
Glass fiber type	E type or ECR type
Resin type	Vinylester resin
Fiber weight content (%)	≥75
Cure ratio (%)	≥98
T_g (°C)	≥110
Moisture uptake (%)	≤0.15
Direct shear strength (MPa)	≥170
Short-beam shear strength (MPa)	≥59
Four-point flexural strength (MPa)	Not specified
Flexural modulus of elasticity (GPa)	≥50

respectively, after 180 days of immersion in an alkaline solution at 60°C.

5. The microstructural observations revealed no significant damage to the internal microstructure or at the fiber–matrix interface in the VE-based GFRP dowels. No changes in the physical properties or significant chemical degradation were observed, even near the dowel surface.
6. The UP-based GFRP dowels were significantly affected by aging in a concrete environment. Strength retentions of 75 and 65% were measured in transverse shear strength and flexural strength, respectively, after 180 days of immersion in an alkaline solution at 60°C.
7. The degradation of the UP-based GFRP dowels was confirmed by a loss of T_g of 6% (5°C), showing a slight irreversible degradation of the matrix. The microstructure of the UP-based GFRP dowel confirms these results by way of local delamination and degraded surfaces. Delamination and fiber debonding also occurred near the surface of the GFRP dowels, leading to an increase in the void content. The significant losses of properties could be also explained by a plasticization phenomenon in the matrix.
8. According to the long-term predictions, the transverse shear-strength retention of the VE-based GFRP dowels aged in an alkaline solution would decrease by 10% after 100 years at 10 and 25°C, which indicates the high stability of the VE-based GFRP dowels in a concrete environment.
9. According to the long-term predictions, the transverse shear-strength retention of the UP-based GFRP dowels aged in an alkaline solution would decrease by 26% after 100 years at 10°C.
10. This study clearly showed that the durability of the VE-based GFRP dowels tested was not significantly affected by immersion in an alkaline solution for 180 days at 60°C, which is harsher than an actual concrete environment. Finally, based on the results of this research study, the MTQ has issued new material specification for GFRP dowels (Table 6) to ensure product performance and longevity. This will accelerate the use of these GFRP dowels in jointed concrete pavements in the province of Québec.

Acknowledgments

The authors thank the Ministry of Transportation of Québec, the Natural Sciences and Engineering Research Council of Canada, the Fonds québécois de la recherche sur la nature et les technologies for their financial support, and Pultrall for the donation of the GFRP dowels. The authors would like to thank the technical staff in the material and structural laboratory at the Department of Civil

Engineering, University of Sherbrooke for their assistance in testing the specimens.

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